Table of Basic Integrals

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Simple Integrals

(1) $\int 0\,dx = 0$

(4) $\int x^n\,dx = \frac{x^{n+1}}{n+1}$

(7) $\int \frac{1}{x^n}\,dx = \frac{-1}{(n-1)x^{n-1}}$

(10) $\int \frac{1}{x}\,dx = \ln|x|$

(13) $\int a^x\,dx = \frac{a^x}{\ln a} \quad (0 < a \neq 1)$

General Integrals

(2) $\int k\,dx = kx \quad (k = \text{constant})$

(5) $\int (ax+b)^n\,dx = \frac{1}{a}\frac{(ax+b)^{n+1}}{(n+1)}$

(8) $\int \frac{1}{(ax+b)^n}\,dx = \frac{-1}{a(n-1)(ax+b)^{n-1}}$

(11) $\int \frac{1}{(ax+b)}\,dx = \frac{1}{a}\ln|ax+b|$

(14) $\int e^{ax+b}\,dx = \frac{1}{a}e^{ax+b}$

Change of Variables

(3) $\int k\,u(x)du = k\int u(x)du$

(6) $\int u(x)^n du = \frac{u(x)^{n+1}}{n+1}$

(9) $\int \frac{du}{u(x)^n} = \frac{-1}{(n-1)u(x)^{n-1}}$

(12) $\int \frac{du}{u(x)} = \ln|u(x)|$

(15) $\int e^{u(x)}du = e^{u(x)}$

Simple trigonometric Integrals

(16) $\int \sin x\,dx = -\cos x$

(19) $\int \cos x\,dx = \sin x$

(22) $\int \tan x\,dx = -\ln|\cos x|$

(25) $\int \cot x\,dx = \ln|\sin x|$

General trigonometric Integrals

(17) $\int \sin(ax+b)\,dx = -\frac{1}{a}\cos(ax+b)$

(20) $\int \cos(ax+b)\,dx = \frac{1}{a}\sin(ax+b)$

(23) $\int \tan(ax+b)\,dx = -\frac{1}{a}\ln|\cos(ax+b)|$

(26) $\int \cot(ax+b)\,dx = \frac{1}{a}\ln|\sin(ax+b)|$

Change of Variables

(18) $\int \sin(u(x))du = -\cos(u(x))$

(21) $\int \cos(u(x))du = \sin(u(x))$

(24) $\int \tan(u(x))du = -\ln|\cos(u(x))|$

(27) $\int \cot(u(x))du = \ln|\sin(u(x))|$

Rational and Irrational Integrals

(28) $\int \frac{dx}{a^2-x^2} = \frac{1}{2a}\ln\left|\frac{x+a}{x-a}\right|$

(33) $\int \frac{dx}{\sqrt{a^2-x^2}} = \sin^{-1}\frac{x}{a}$

(36) $\int \frac{dx}{\sqrt{a^2+x^2}} = \ln\left(\frac{x}{a} + \sqrt{\left(\frac{x}{a}\right)^2+1}\right)$

(39) $\int \frac{dx}{\sqrt{x^2-a^2}} = \ln\left(\frac{x}{a} - \sqrt{\left(\frac{x}{a}\right)^2-1}\right)$

Rational and Irrational Integrals

(29) $\int \frac{dx}{a^2-x^2} = \frac{1}{a}\tanh^{-1}\frac{x}{a}$

(31) $\int \frac{dx}{a^2+x^2} = \frac{1}{a}\tan^{-1}\frac{x}{a}$

(34) $\int \frac{dx}{\sqrt{a^2-x^2}} = -\cos^{-1}\frac{x}{a}$

(37) $\int \frac{dx}{\sqrt{a^2+x^2}} = \sinh^{-1}\frac{x}{a}$

(40) $\int \frac{dx}{\sqrt{x^2-a^2}} = \cosh^{-1}\frac{x}{a}$

Change of Variables

(30) $\int \frac{du}{a^2-u(x)^2} = \frac{1}{a}\tanh^{-1}\frac{u(x)}{a}$

(32) $\int \frac{du}{a^2+u(x)^2} = \frac{1}{a}\tan^{-1}\frac{u(x)}{a}$

(35) $\int \frac{du}{\sqrt{a^2-u(x)^2}} = \sin^{-1}\frac{u(x)}{a}$

(38) $\int \frac{dx}{\sqrt{a^2+u(x)^2}} = \sinh^{-1}\frac{u(x)}{a}$

(41) $\int \frac{du}{\sqrt{u(x)^2-a^2}} = \cosh^{-1}\frac{u(x)}{a}$

Hyperbolic Trigonometric Integrals

(42) $\int \sinh x\,dx = \cosh x$

(45) $\int \frac{dx}{\sinh x} = \ln\left|\tanh\left(\frac{x}{2}\right)\right|$

(43) $\int \cosh x\,dx = \sinh x$

(46) $\int \frac{dx}{\cosh x} = 2\tan^{-1}(e^x)$

(44) $\int \tanh x = \ln(\cosh x)$

(47) $\int \frac{dx}{\tanh x} = \ln(\sinh x)$

- Integral By Parts $\int u\,dv = u\,v - \int v\,du$

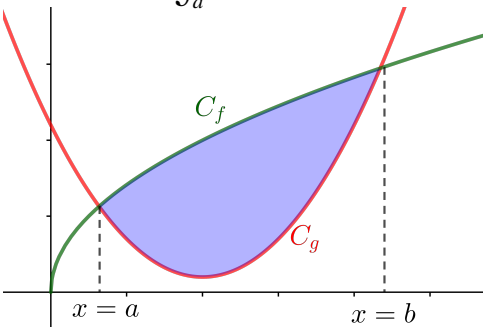
(48)	$\int e^{ax}\sin(bx)\,dx = \frac{e^{ax}}{a^2+b^2}\Big(a\sin(bx) - b\cos(bx)\Big)$	(49)	$\int e^{ax}\cos(bx)\,dx = \frac{e^{ax}}{a^2+b^2}\Big(a\cos(bx) + b\sin(bx)\Big)$
(50)	$\int \ln x\,dx = x\ln x - x$	(51)	$\int p(x)e^{ax}\,dx = \frac{e^{ax}}{a}\left(p(x) - \frac{p'(x)}{a} + \cdots + (-1)^n\frac{p^{(n)}(x)}{a^n}\right)$

(52) $\int p(x)e^{ax}\,dx = \frac{e^{ax}}{a}\left(p(x) - \frac{p'(x)}{a} + \frac{p''(x)}{a^2} + \cdots + (-1)^n\frac{p^{(n)}(x)}{a^n}\right)$

• Area bounded by

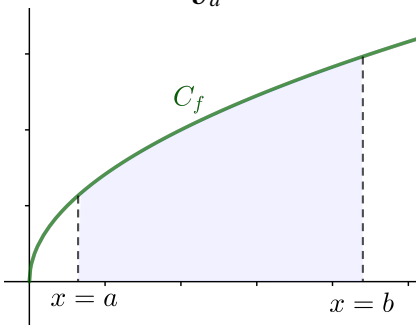
(C_f), (C_g), the lines $x = a$ and $x = b$

Area $= \int_a^b \Big[f(x) - g(x)\Big]dx$



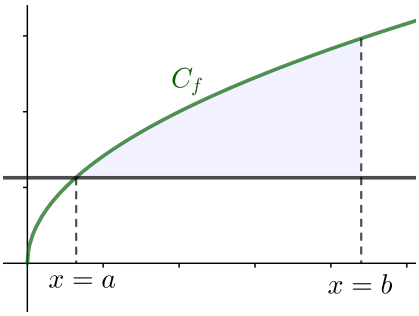
(C_f), y axis, the lines $x = a$ and $x = b$

Area $= \int_a^b f(x)dx$



(C_f), $y = m$, the lines $x = a$ and $x = b$

Area $= \int_a^b \Big[f(x) - y\Big]dx$



• First fundamental theorem of integral

(53) $F(x) = \int f(x)\,dx \Longleftrightarrow F'(x) = f(x)$ (54) $\int_a^b f(x)\,dx = \Big[F(x)\Big]_a^b = F(b) - F(a)$

• Second fundamental theorem of integral

(55) $F(x) = \int_a^{u(x)} h(t)\,dt \Longrightarrow F'(x) = \frac{d}{dx}F(x) = f\Big(u(x)\Big)\cdot u'(x)$